



Dermatology

Workbook



Dr Manish Soni
MBBS, MD (Gold Medalist)
IADVL Fellow (AIIMS).



INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.



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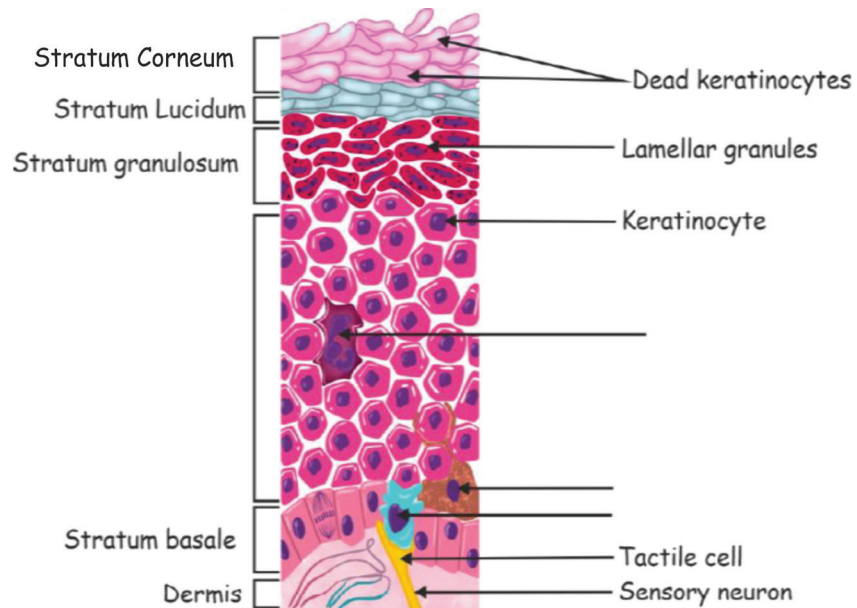
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LAYERS OF SKIN

Active space

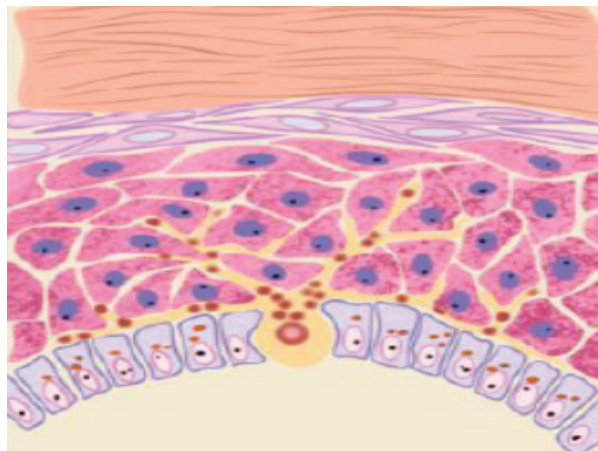
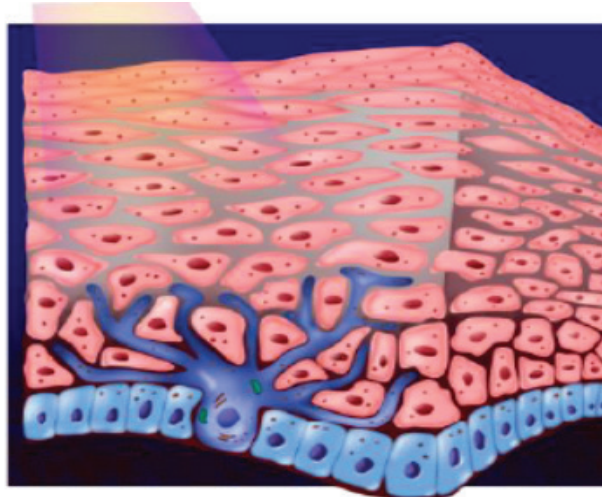
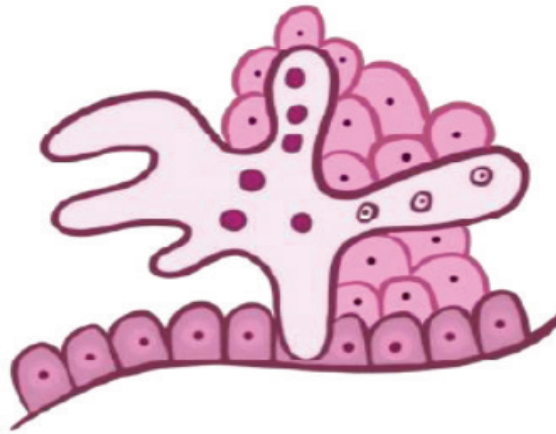
MIST



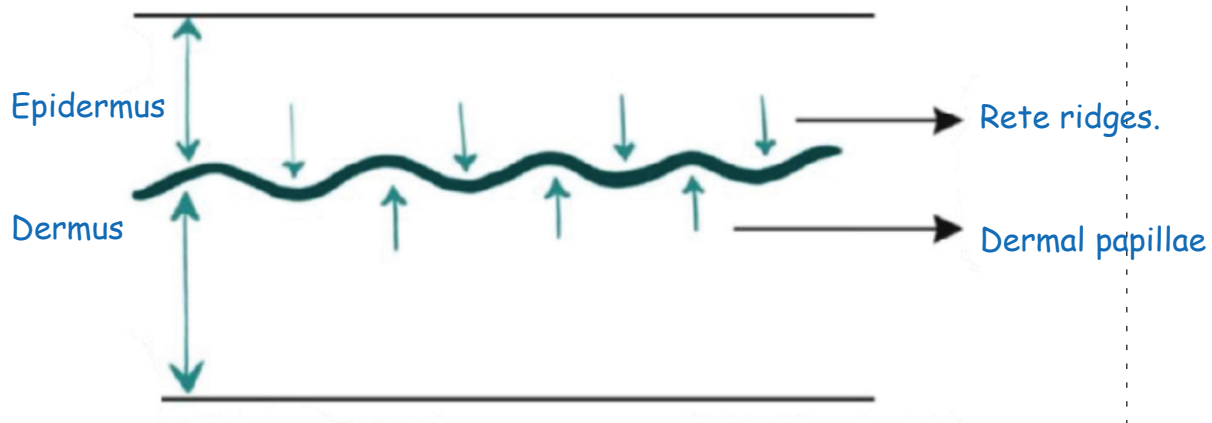
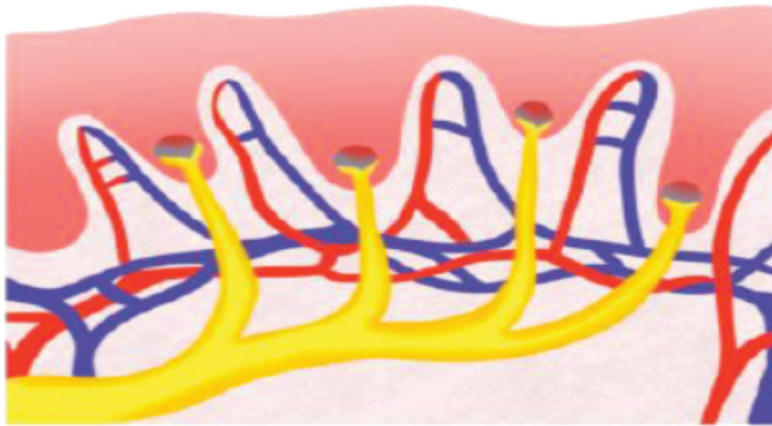
CELLS OF EPIDERMIS

1. Keratinocytes

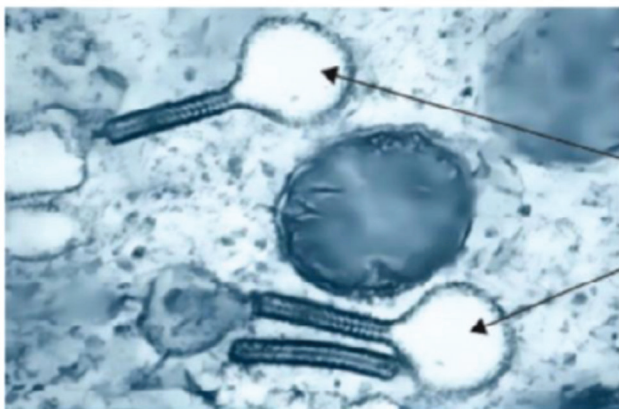
2. Melanocytes :



3. Merkel Cells :



4. Langerhans Cells :



Active space

Important facts

1. Markers of melanocytes →

2. Markers of Langerhans cells →

↑ number of Langerhans cells are seen in

↓ number of Langerhans cells are seen in

Langerhans cells histiocytosis present with:

-
-
-

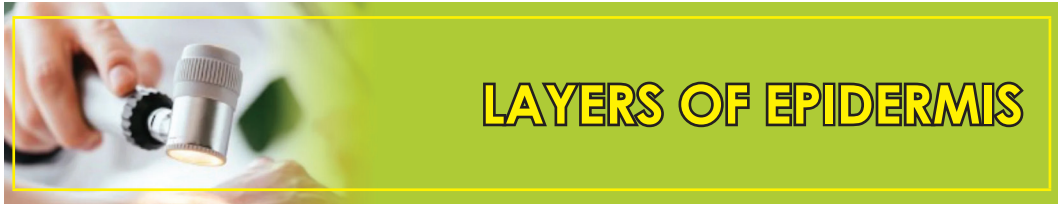
Seborrheic Dermatitis:

✱ They are skin inflammation in

✱ Seborrheic areas are areas where the sebaceous gland is present in excess:

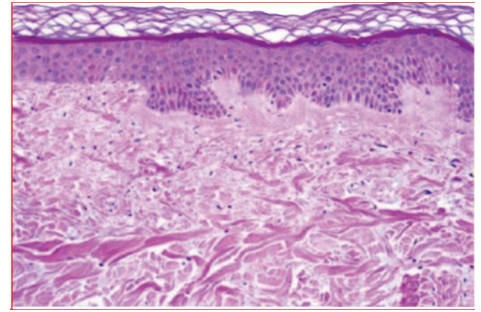
Cells of Epidermis	Derived from
1. Keratinocyte	•
2. Melanocytes	•
3. Merkel cells	•
4. Langerhans Cells	•

Active space



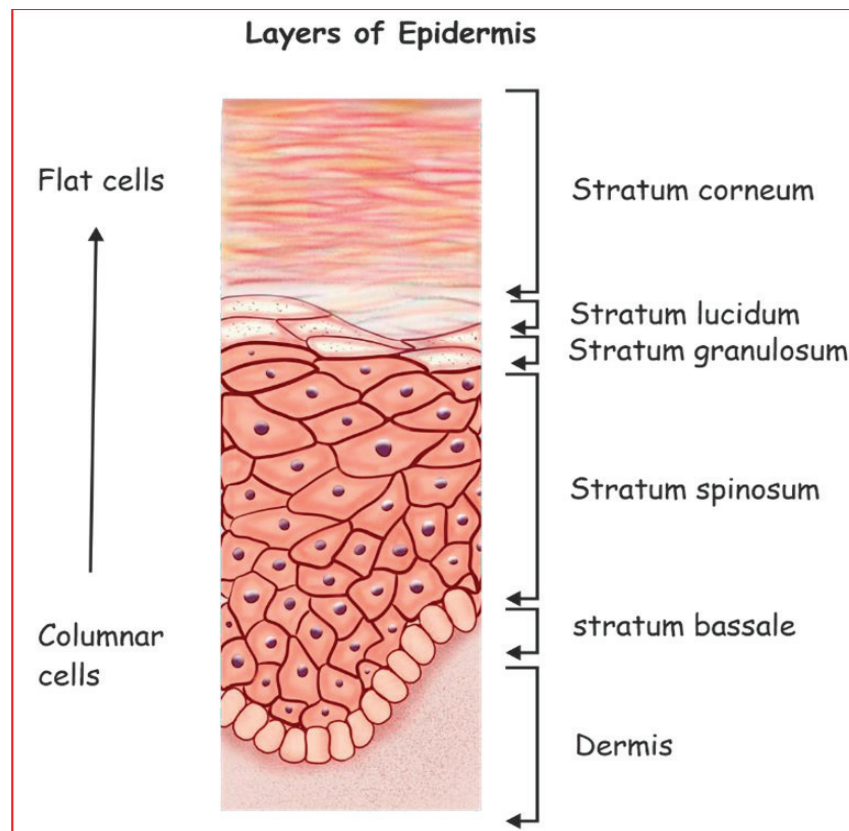
LAYERS OF EPIDERMIS

- STRATUM BASALE
- STRATUM SPINOSUM
- STRATUM GRANULOSUM
- STRATUM LUCIDUM
- STRATUM CORNEUM



Normal skin layer under microscope

DIAGRAMMATIC REPRESENTATION OF EPIDERMIS



KNOWLEDGE CAPSULE

High Yield Points:

- ✱ Normal skin epidermal turnover time =
- ✱ Epidermal turnover time in psoriasis is

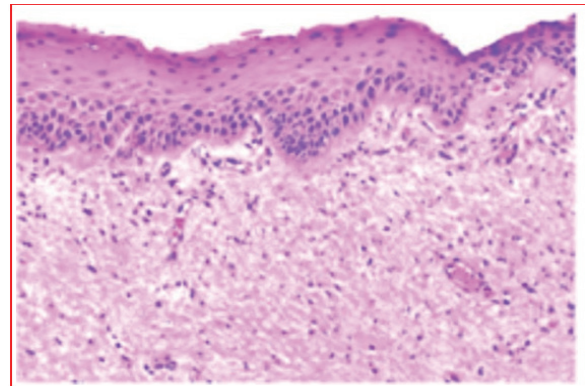
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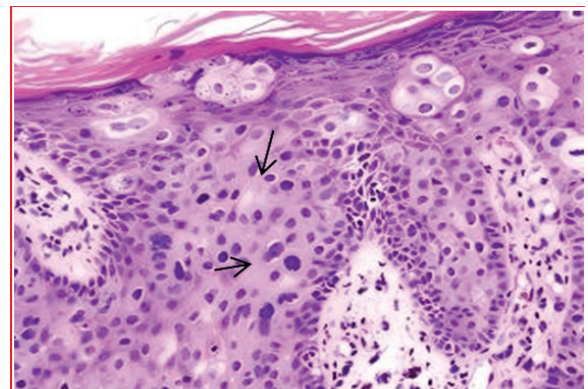


Various Histological Findings which are seen are:

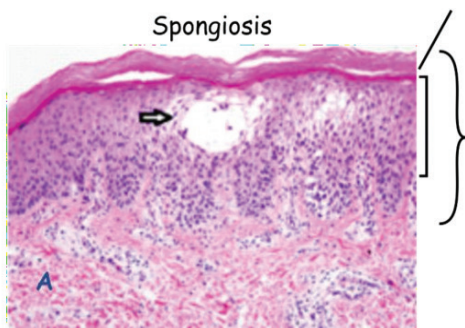
1. Parakeratosis:



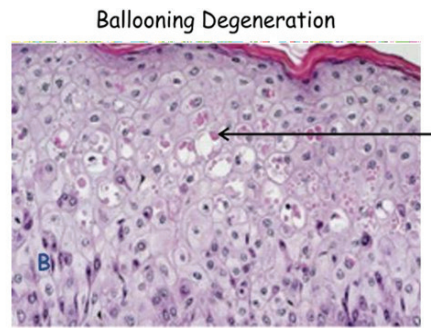
2. Dyskeratosis:



3. Spongiosis:



A. Spongiosis due to extra-cellular accumulation of fluid



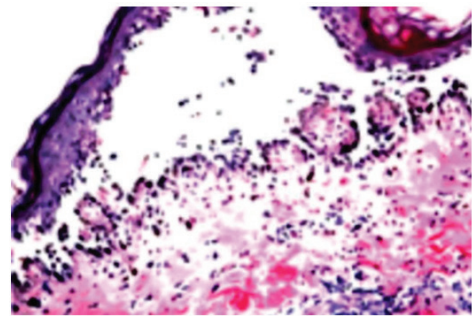
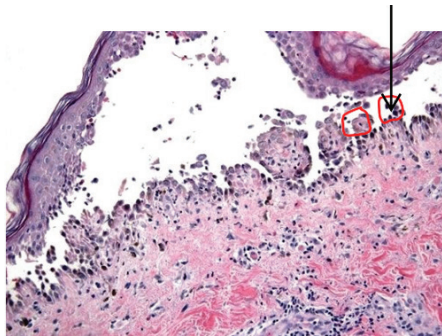
B. Ballooning degeneration due to intra-cellular accumulation of fluid

4. Acanthosis:

5. Agranulosis:

6. Hypo granulosis:

7. Acantholysis:



Acantholysis

8. Micro abscess:

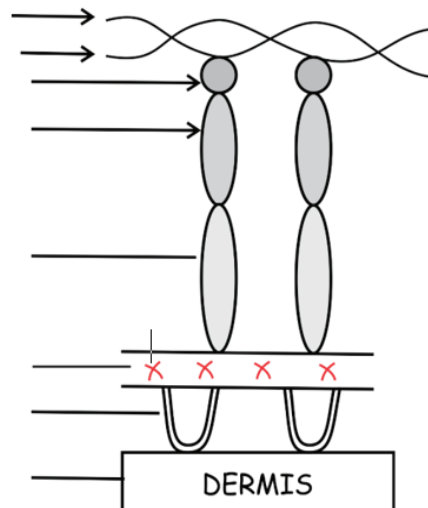
- Small collection of cells is k/a Microabscess

1. Neutrophils	• •
2. Lymphocytes	•
3. Eosinophils	•

BASEMENT MEMBRANE ZONE, DERMIS & SUBCUTANEOUS LAYER

II. BASEMENT MEMBRANE ZONE

Basement Membrane Zone/Dermoepidermal Junction:



Sub-epidermal Disorders:

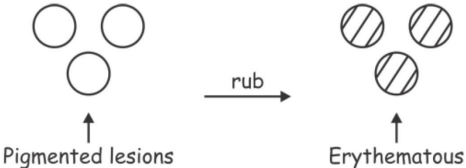
	Disorders Q	Defective protein Q
1.	Bullous Pemphigoid	
2.	Linear IgA Disease	
3.	Lichen Planus Pemphigoid	
4.	Herpes Gestationis	
5.	Epidermolysis Bullosa Junctional	
6.	Epidermolysis Bullosa Dystrophicans	

III. Dermis:

Urticaria & Urticaria Pigmentosa:

KNOWLEDGE CAPSULE

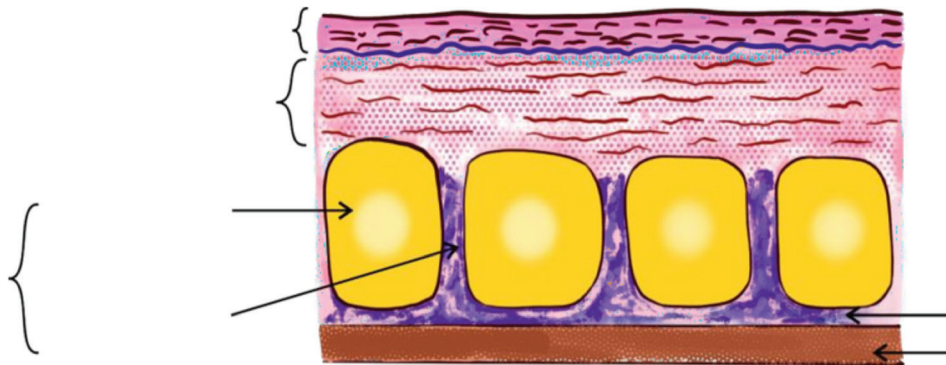
Important facts:
Disorders of Mast Cell

	Urticaria		Urticaria Pigmentosa
1.		1.	
2.		2.	 The diagram illustrates the Darier sign. On the left, three white circles represent 'Pigmented lesions'. An upward arrow points from this label to the circles. A horizontal arrow labeled 'rub' points to the right. On the right, three circles with diagonal hatching represent 'Erythematous' lesions. An upward arrow points from this label to the hatched circles.
3.	Triple response of Lewis: erythema → flare → wheal	3.	
4.			



Darier sign

IV. Subcutaneous Layer/Hypodermis/Panniculus:



Panniculitis:

Inflammation of the panniculus layer



Active space

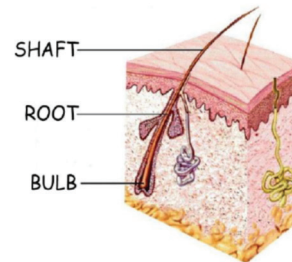
Behcet's Disease:



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HAIR FOLLICLES



HAIR CYCLE

ANAGEN

CATAGEN

TELOGEN



HAIR DISORDER

ANAGEN EFFLUVIUM

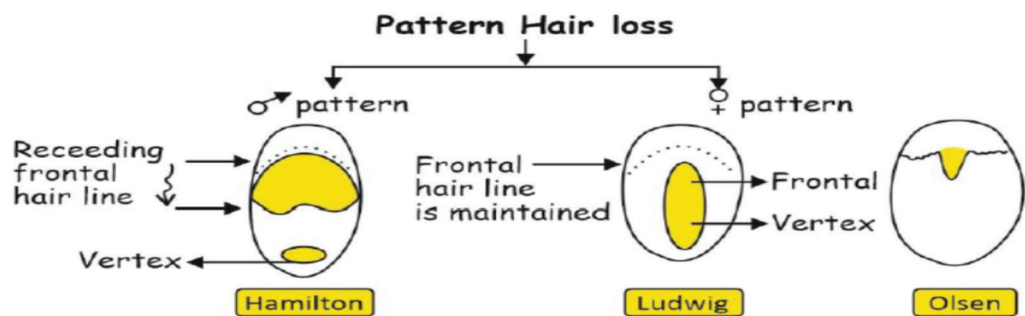
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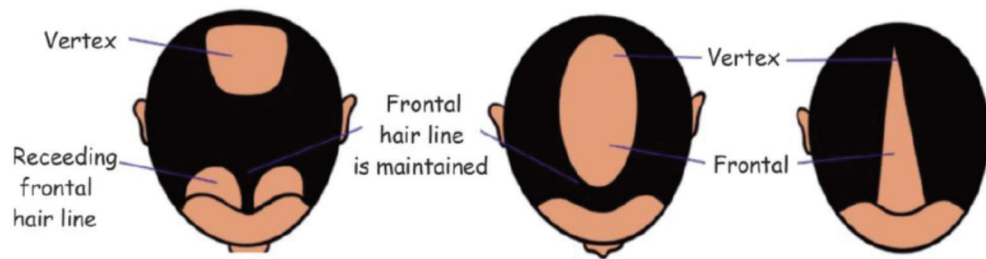
TELOGEN EFFLUVIUM

ANDROGENETIC ALOPECIA:



Clinical Features - Pattern hair loss





ALOPECIA AREATA:



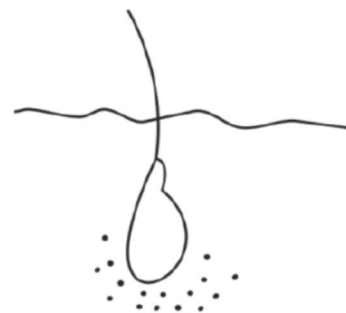
Active space

MIST

Autoimmune disorder, autoreactive CD 8 T - Cells immune response to hair follicle associated antigen.

Clinical Features:

Nails:



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Trichotillomania:

ALOPECIA

1. Scarring (permanent) - Irreversible/ Cicatricial
2. Non-scarring (temporary) - Reversible/Non-Cicatricial

Non-scarring alopecia:

Primary:

1. Anagen effluvium
2. Telogen effluvium
3. Trichotillomania
4. Alopecia areata
5. Androgenetic alopecia
6. Tractional alopecia/Hair style alopecia
7. Tinea Capitis (non-inflammation)

Secondary

1. SLE
2. Secondary syphilis
3. Hypothyroidism
4. Hyperthyroidism
5. Hypopituitarism
6. Deficiency of zinc, iron and biotin

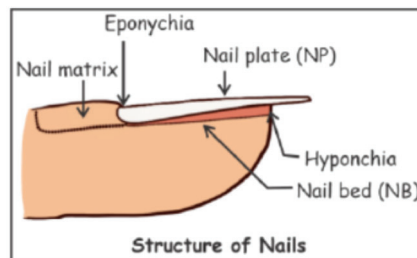
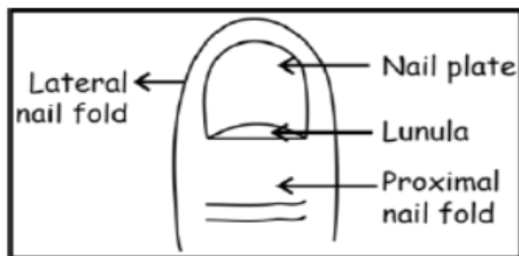


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STRUCTURE OF NAIL & NAIL DISORDER

NAILS

Structure



Nail changes seen in Psoriasis:

1. Pitting -
2. Oil drop sign -
3. Leukonychia
4. Onycholysis
5. Subungual Hyperkeratosis - Nail bed thickening

- Leukonychia
- Onycholysis
- Subungual Hyperkeratosis

} Also seen in 'Onychomycosis'

Nail Changes Seen In Lichen Planus:

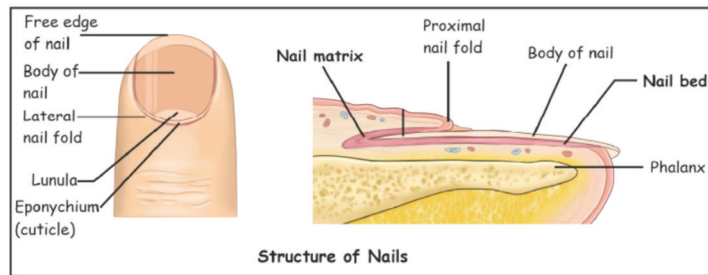
Pterygium -



Pterygium

Winged-shaped extension of the proximal nail

Structure of Nails:



- **Mees' lines of Aldrich:**

Mees' lines appear after an episode of poisoning with arsenic, thallium



Mees's lines of Aldrich

- **Half and Half nails:** Also known as Lindsay's nails show the proximal portion of the nail white

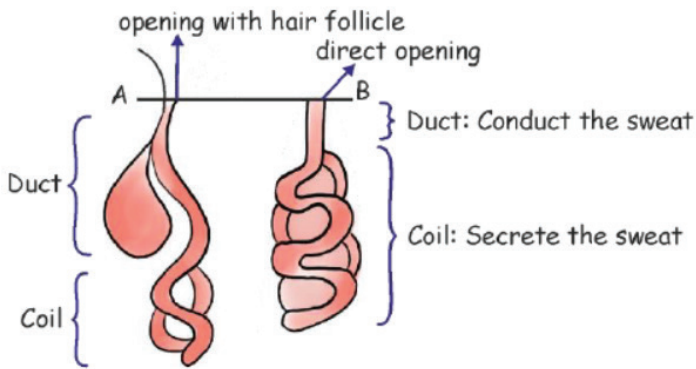


Half & Half nails

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SWEAT GLANDS



A-Apocrine glands; B-Eccrine glands

Difference Between Eccrine & Apocrine Glands:

Feature	Eccrine	Apocrine
1. Duct opens		
2. Site		
3. Nerve innervation		
4. Disorders		

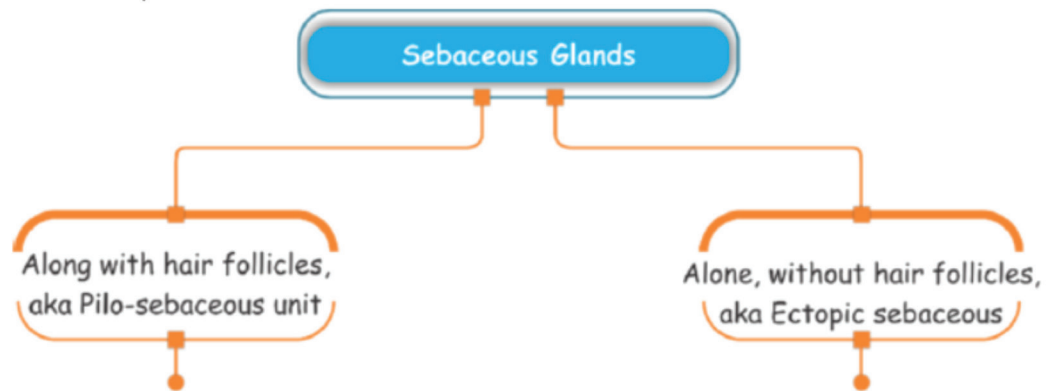
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Disorders Related to Eccrine & Apocrine:

Feature	Eccrine	Apocrine
Disorders	i. Anhidrosis ii. Hyperhidrosis iii. Miliaria^Q	i. Bromhidrosis: ii. Chromhidrosis: iii. Fox Fordyce's Disease :^Q iv. Hiradenitis Suppurativa:

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SEBACEOUS GLANDS



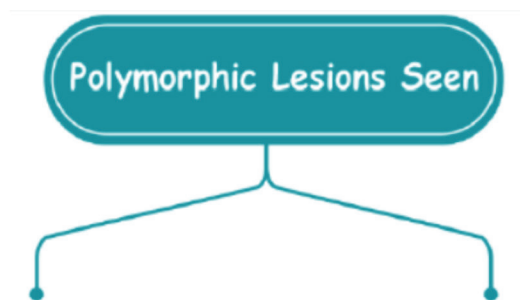
Rosacea/Acne Rosacea:

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ACNE VULGARIS

CLINICAL FEATURES

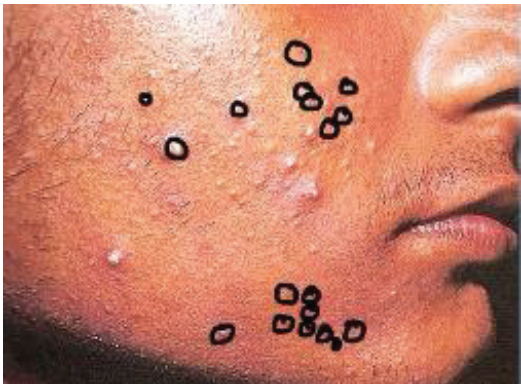


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CLINICAL GRADING & TREATMENT

Grades & Treatment of Acne Vulgaris:

Grade	Lesions	Treatment
I	Mainly comedons	
II	Mainly papules	
III	Mainly pustules	
IV (severe acne) ^Q	Nodules, cyst (Nodulocystic) ^Q	



Active space



Active space

MIST

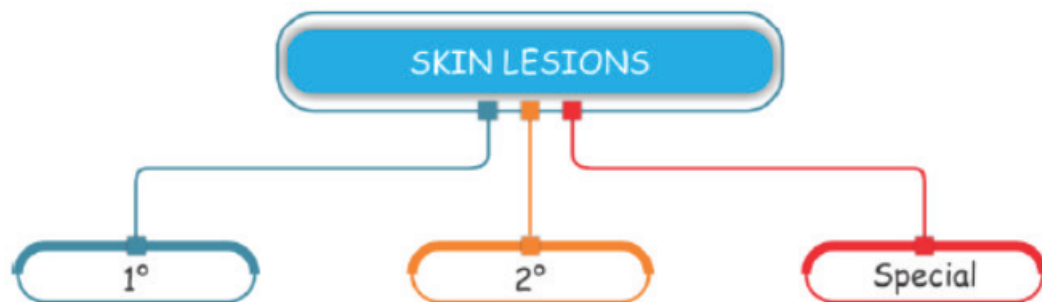
Active space

MIST



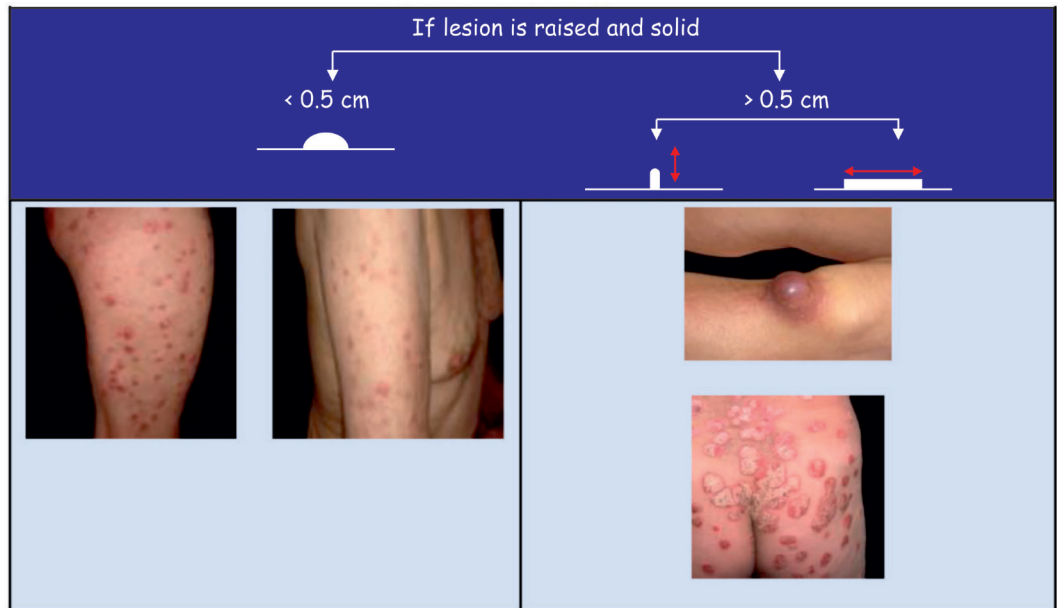
Skin lesions: Divided into Three Types:

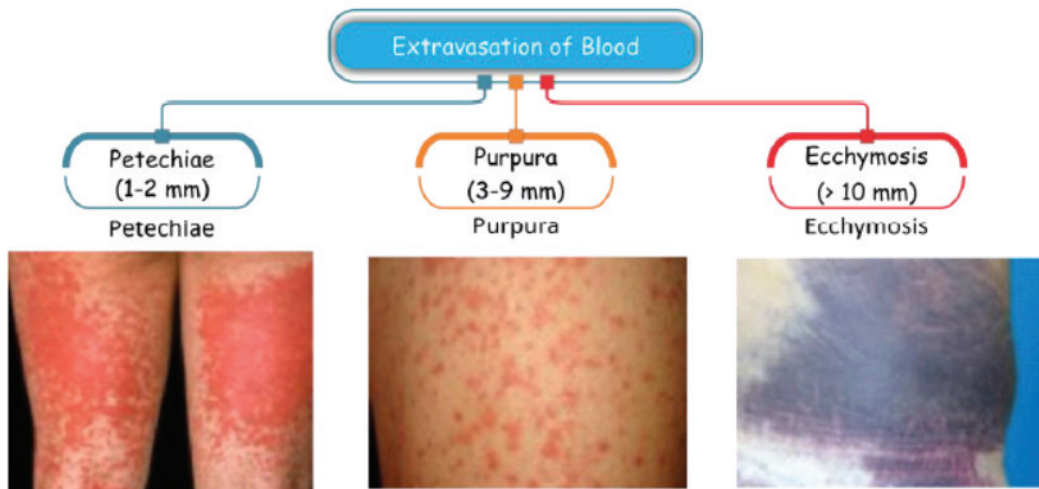
1. Primary skin lesion
2. Secondary skin lesion
3. Special skin lesion



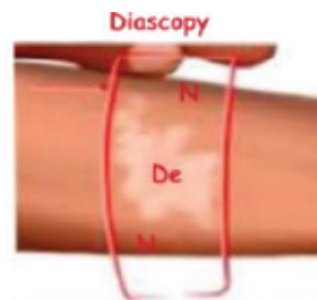
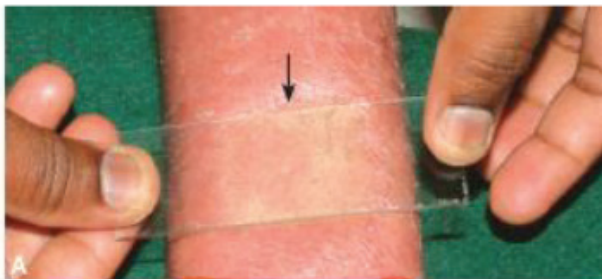
Primary Skin Lesions:







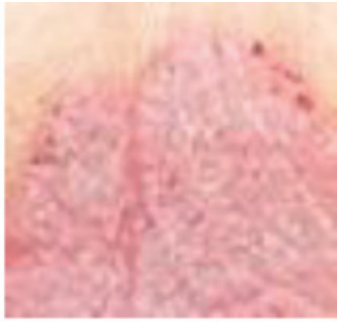
Diascopy:



2° Skin Lesions:



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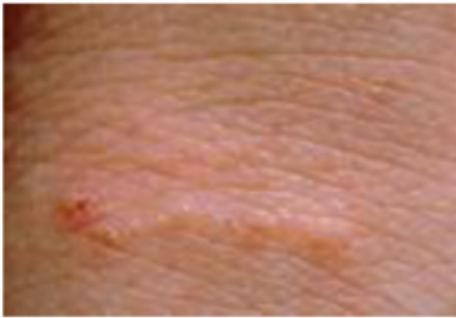


Lichenification:



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Special Skin Lesions:



Transient -



Types of Scales:

i. Ichthyosis:



ii. Pityriasis Rosea:



iii. Rupoid scales:



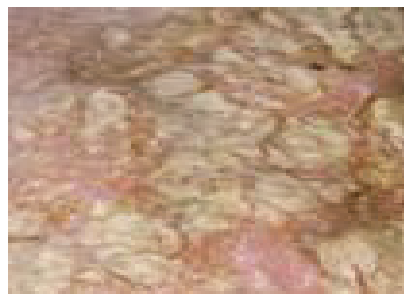
iv. Silvery white scales:



v. Branny scales



vi. Cradle Cap:



Pattern and Arrangement of Lesions:

1. Dermatomal lesions:



2. Annular lesions:



3. Serpiginous lesions:



4. Arcuate and polycyclic lesions:



Distribution of Lesions:

1. Acral:



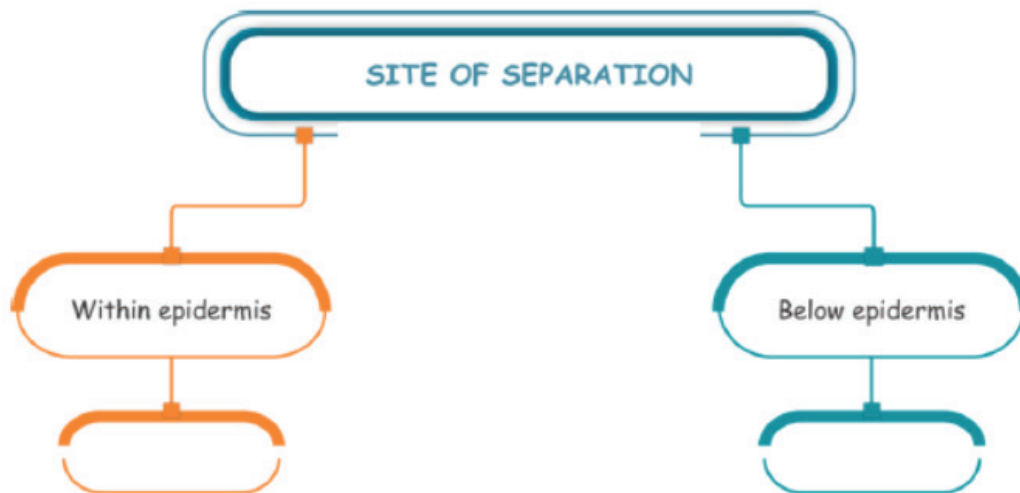
2. Herpetiform:



3. Intertriginous



Active space

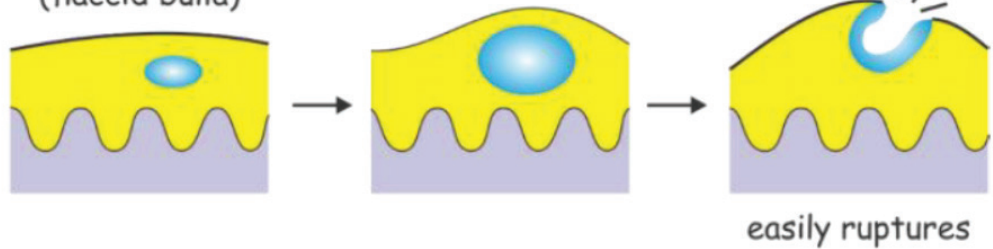


(1) Pemphigus

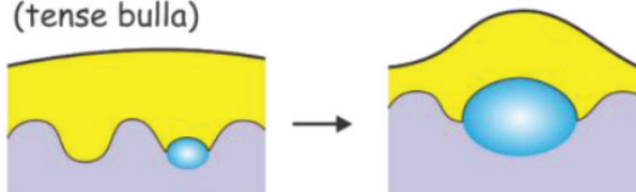


Pemphigus-erosions with crusting seen

Intraepidermal blister ——— e.g., pemphigus
(flaccid bulla)

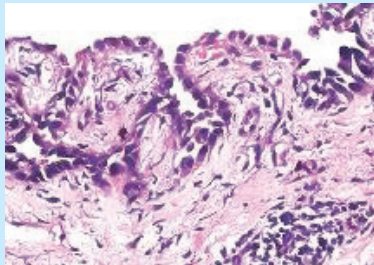


Subepidermal blister ——— e.g., bullous pemphigoid
(tense bulla)

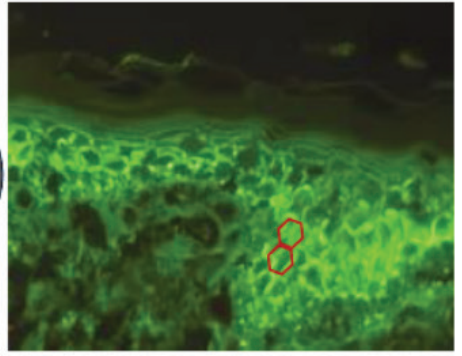
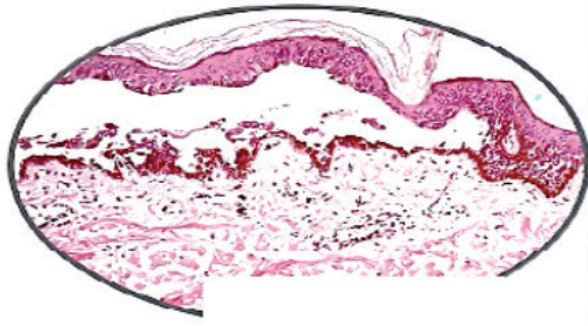


Because a subepidermal blister is caused with all layers of the epidermis intact, it does not easily rupture

	Type of Pemphigus	Defect	Site of separation	Bullae	Easily rupture
1.	Pemphigus foliaceus				
2.	Pemphigus vulgaris				

	Pemphigus Vulgaris	Pemphigus foliaceus
1. Defect:		
2. C/F: Buccal mucosa involvement: Skin: Sign: Nikolsky sign:		
3. Investigations: Histology - (a) Acanthocytes:		
(b)		
DIF (direct immunofluorescence)		

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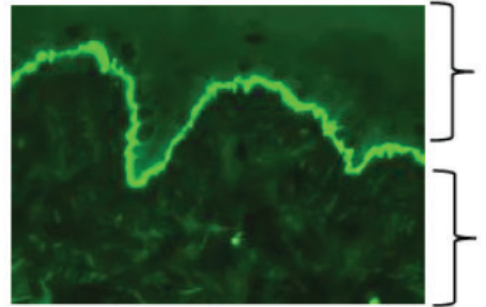


Treatment

(2) BULLOUS PEMPHIGOID



A.



B.

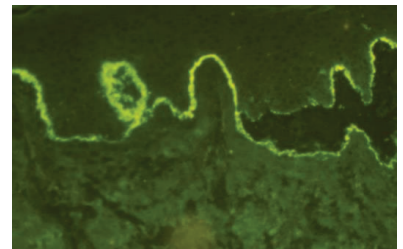
Active space



Pemphigus		Pemphigoid
1. Age of onset 2. Blisters		
8. DIF or Direct Immunofluorescence:		
9. Histology:		

Active space

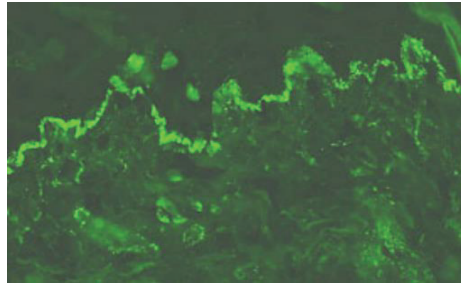
(3) LINEAR IGA DISEASE



(4) DERMATITIS HERPETIFORMIS



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Active space

MIST

(5) HERPES GESTATIONIS



(6) EPIDERMOLYSIS BULLOSA / MECHANOBULLOUS DISORDERS

Congenital
Congenital
Early onset
Genetic defect
3 subtypes with genetic defects: (a) Epidermolysis Bullosa simplex (b) Epidermolysis Bullosa junctional (c) Epidermolysis Bullosa dystrophicans

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	7. Darier Disease	8. Hailey - Hailey disease
Inheritance		
Defect: Ca^{2+} ATPase		
Gene:		
Clinical Features:		
Site -		
		

(9) ERYTHEMA MULTIFORME



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(10) STEVEN JOHNSON SYNDROME & TOXIC EPIDERMAL NECROLYSIS
(SJS) (TEN)





PIGMENTARY DISORDERS



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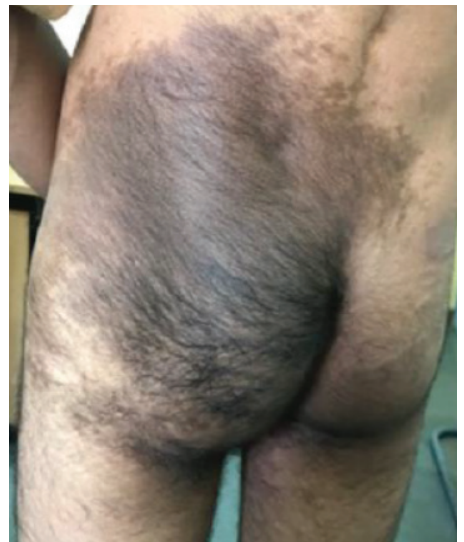
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Active space

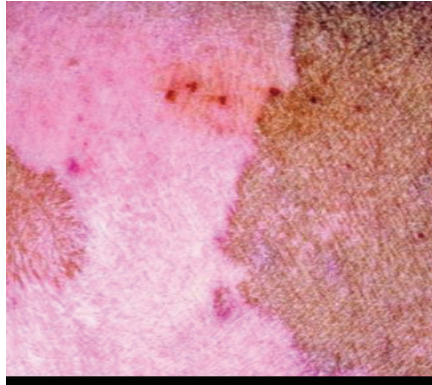
MIST

Beckers nevus:



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Vitiligo:



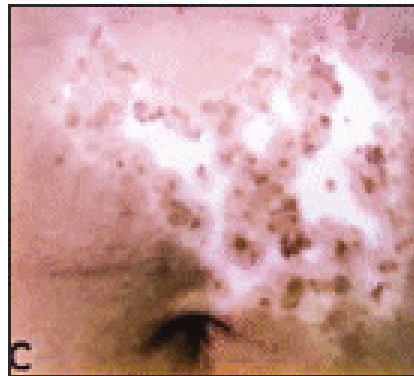
Active space

MIST



Active space

Peibaldism:



Active space

MIST

Contact Leukoderma:



Active space



Pityriasis alba :



Active space

MIST



1. Pitted Keratolysis



2. Erythrasma



3. Trichomycosis
Axillaries

4. Impetigo

	Non-Bullous Impetigo/ Impetigo Contagiosa	Bullous Impetigo
Agent:		
Clinical Features		
Site:		
Complications:		
Treatment:		



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SSSS (Staphylococcal scalded skin syndrome)



Active space

Erysipelas & Cellulitis



	Erysipelas	Cellulitis
Agent		
Borders		
Lymphangitis		
Toxicity		

Active space

Disorders	Colour seen in the woods lamp light
1. Tinea Capitis	
2. Porphyria (Urine)	
3. Pityriasis Versicolor	
4. Erythrasma	
5. Vitiligo	
6. Ash leaf macules (Tuberous Sclerosis)	

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1. PITYRIASIS VERSICOLOR :



2. CANDIDIASIS :



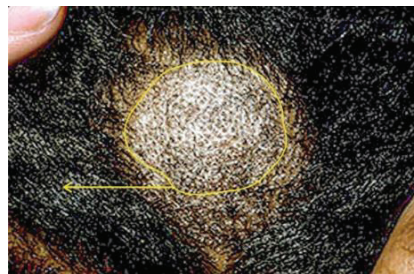
Disease	Type of Vaginal discharge	Treatment
Candidiasis		
Trichomoniasis		
Bacterial Vaginosis		

Active space

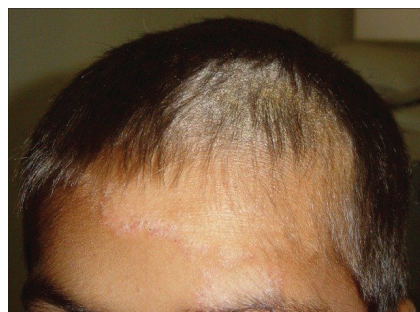
3. DERMATOPHYTOSIS/TINEA

Agents	Skin	Hair	Nail
Trichophyton			
Epidermophyton			
Microsporum			

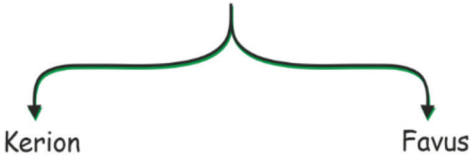
Active space



Tinea Capitis:



Active space

Inflammation (Scarring alopecia)		Non-Inflammation (Non-Scarring alopecia)	
 <pre>graph TD; A["Inflammation (Scarring alopecia)"] --> B["Kerion"]; A --> C["Favus"];</pre>		 <pre>graph TD; A["Non-Inflammation (Non-Scarring alopecia)"] --> B["Grey patch"]; A --> C["Black dot"];</pre>	

Investigation: Wood's lamp examination, KOH mount

Active space

4. SPOROTRICHOSIS OR ROSE GARDENER'S DISEASE

	Variant	Clinical Features	DOC
1.	Lymphatic Most Common		

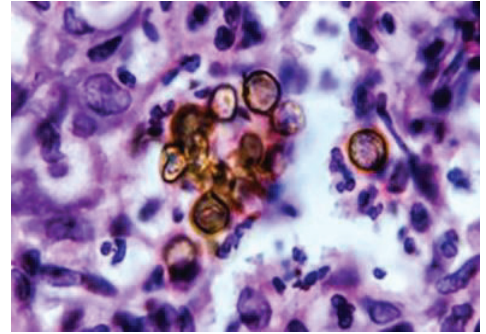
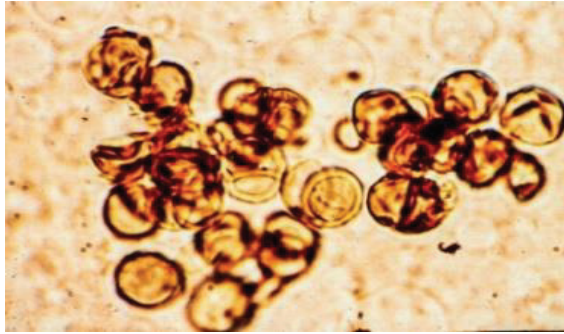


5. CHROMOBLASTOMYCOSIS



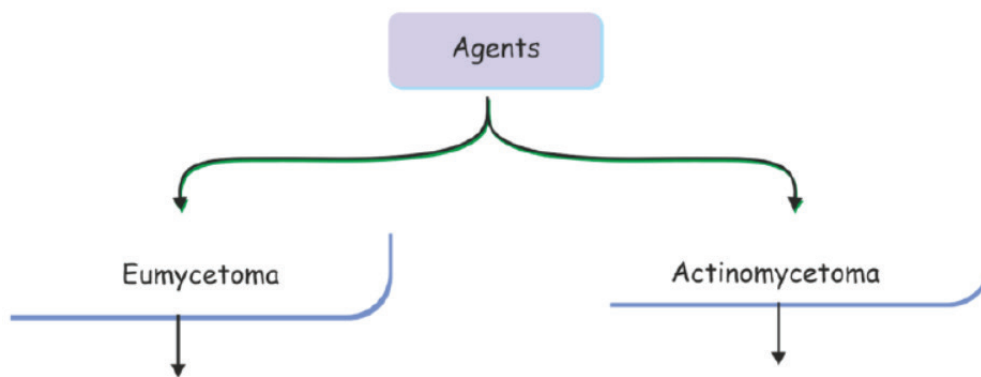
Active space

Histology:



Active space

7. MADURA FOOT/MYCETOMA/MADURAMYCOSIS:





MYCOBACTERIAL INFECTIONS



CUTANEOUS TUBERCULOSIS/SKIN TB

1. Tubercular Chancre:



2. TBVC - Tuberculosis Verrucosa Cutis



3. Lupus Vulgaris



Active space

Diascopy :- Apple jelly nodules

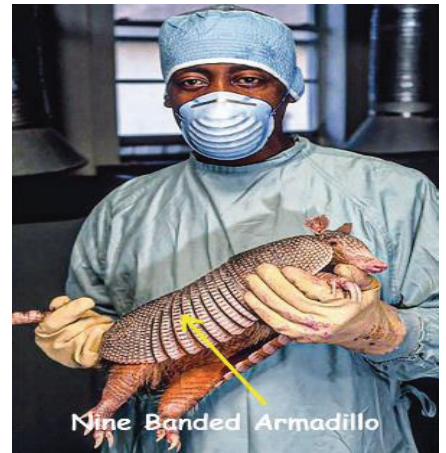


4. Scrofuloderma:



Active space

LEPROSY



Active space

Most Common type of leprosy →

Rarest type of leprosy →

MC Cranial nerve involves in leprosy →

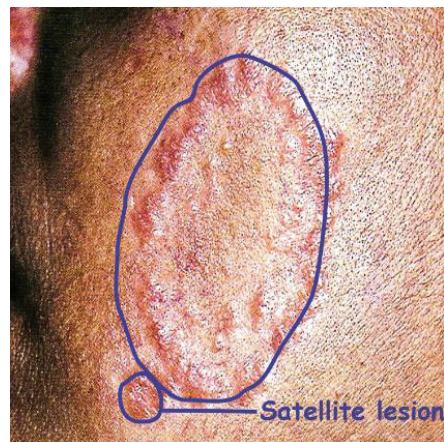
MC peripheral nerve leprosy →

1st sensation lost →

Satellite lesion →

Punched out lesion →

Virulence of *Mycobacterium Leprae* →



Active space



WHO CLASSIFICATION

WHO Classification ↓ 3 Cardinal features	2 Types	
	Pauci Bacillary	Multi Bacillary
1. Skin: Hypopigmentation, hypoanaesthetic patches		
2. Nerve: Peripheral nerve thickening		
3. AFB: Skin Slit Smear		

Active space

TREATMENT

Treatment (3 Drugs)		MDT (Multi Drug Therapy)	
		PB (pauci bacillary)	MB (multi bacillary)
1. Rifampicin		⊕	⊕
2. Dapsone		⊕	⊕
3. Clofazimine		⊕	⊕
		⊕	⊕

LEPRA REACTION

Acute exacerbation of pre-existing disease due to change in immunological status of the patient

2 Types:	Type I	Type II
1. Hypersensitivity		
2. Clinical Feature		
3. Treatment		

Active space



1. WARTS



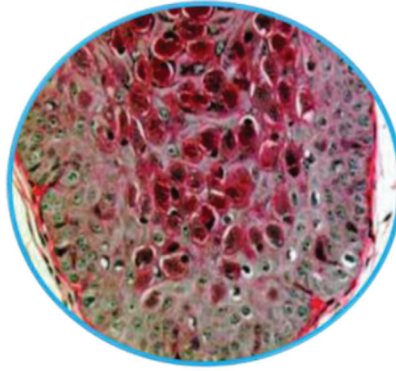
Anogenital Warts

2. MOLLUSCUM CONTAGIOSUM



Active space

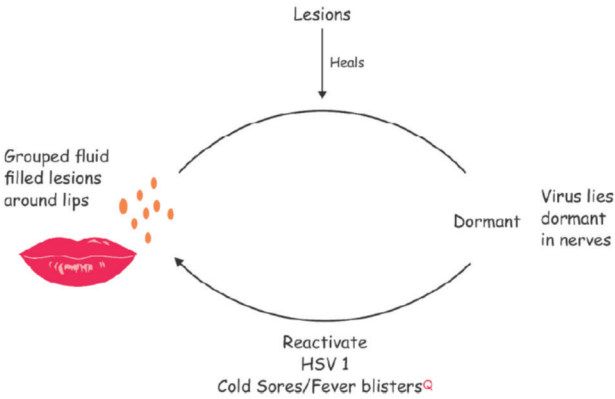
HISTOLOGY & TREATMENT



Active space

HERPES LABIALIS, HERPES GENITALIS & HERPES ZOSTER

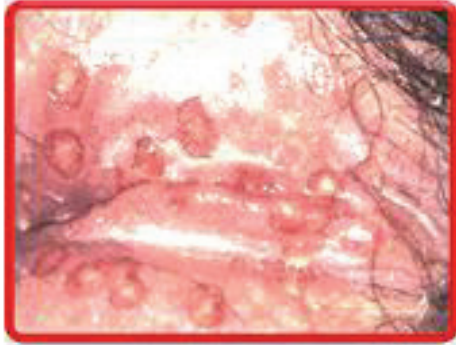
	Herpes labialis	Herpes genitalis (STI)	Herpes zoster
Agent:			
C/F:			



Active space



Active space



Hand, Foot and Mouth Disease

Active space

SEXUALLY TRANSMITTED INFECTIONS (STI)

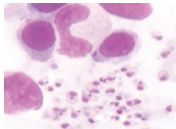
BACTERIAL SEXUALLY TRANSMITTED INFECTION

	1. Syphilis	2. LGV (Lympho Granuloma Venerum)	3. Chancroid	4. Donovano- sis Or Granuloma Inguinale
Agent:				
Incubation period:				
Genital ulcers:				

Active space

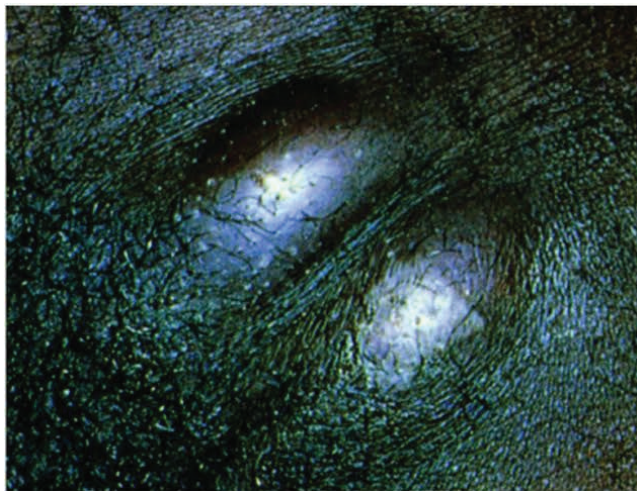
Lymph Node:				
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INVESTIGATION & TREATMENT

Investigation				
1 st DOC:				
2 nd DOC:				

Active space

Benzathine penicillin is the DOC for all types of syphilis except Neuro syphilis. DOC for Neurosyphilis - Aqueous/Crystalline form of penicillin.



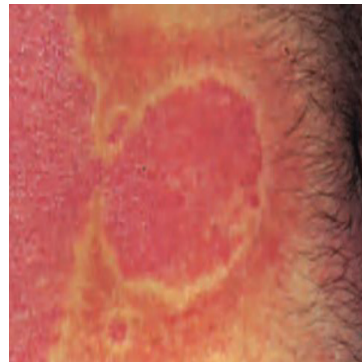
Active space

Active space

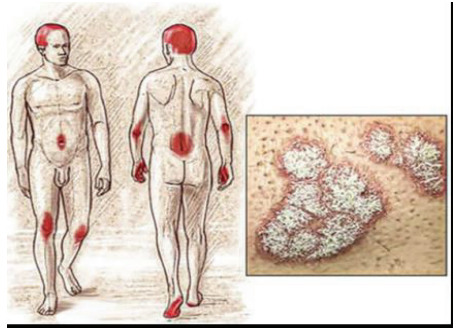
MIST



- ✱ Systemic steroids are C/I in psoriasis because withdrawal can lead to several types like 'Erythrodermic psoriasis' or 'pustular psoriasis'
- ✱ Nails get involved in psoriasis
- ✱ Joint involvement



Active space



Guttate Psoriasis:



Rupioid Psoriasis:



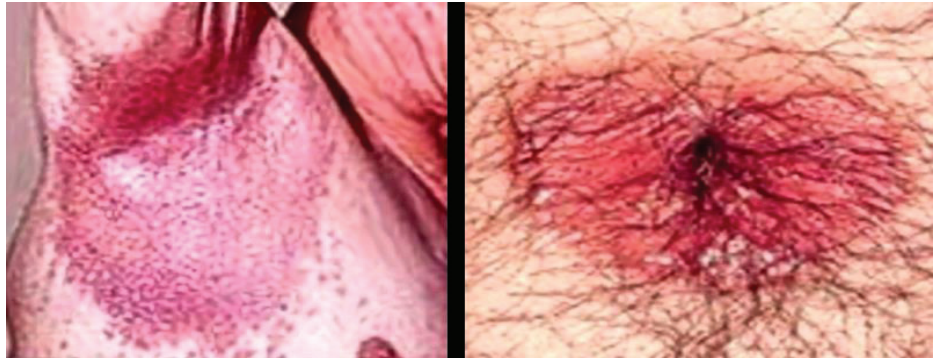
Active space



Scalp Psoriasis:



Inverse Psoriasis:



Psoriatic Arthritis:



Active space

Pustular Psoriasis:



Erythrodermic Psoriasis:

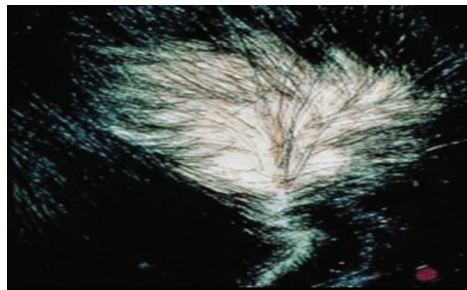


PITYRIASIS ROSEA



Active space

LICHEN PLANUS



Active space

Mnemonic: All 'P's seen in Lichen Planus

- Plain
- Purple
- Polygonal
- Pruritic
- Papules/plaques
- Pattern
- Pterygium
- Pila/hairloss

Active space



1. MALIGNANT MELANOMA



2. BASAL CELL CARCINOMA



3. SQUAMOUS CELL CARCINOMA



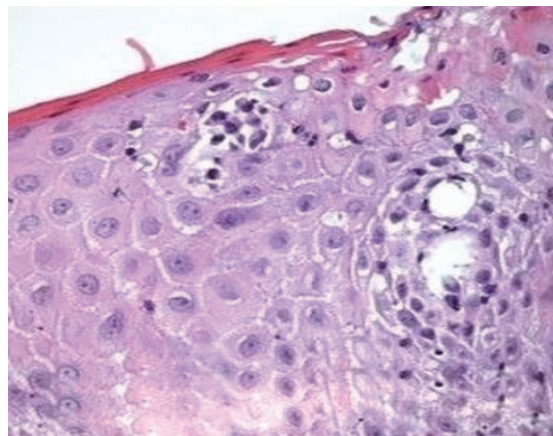
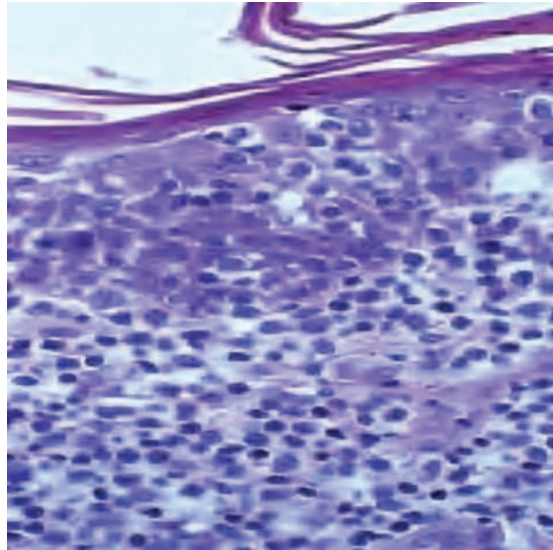
4. MYCOSIS FUNGOIDES



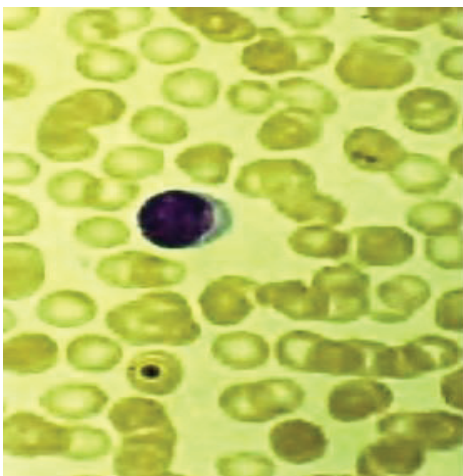
Early MF showing patch



More advanced MF plaques



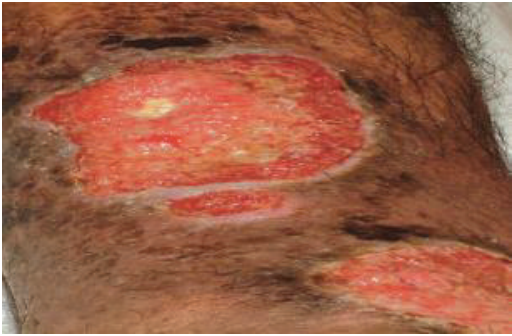
ii.



Active space

MIST

	Sweet Syndrome	Pyoderma Gangrenosum
Clinical Features:		
Associated With		
Doc		



Active space



Active space

SCABIES



Active space

PELLAGRA



Active space

DERMATITIS (ECZEMA)

Active space



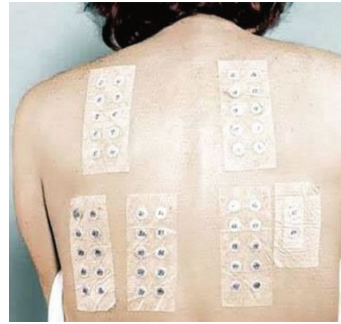
Active space

Atopic Dermatitis:



Active space

Patch Test :



Active space

MIST

Active space

MIST